

DEVELOPMENT RECORD

Skillomate Development Work Report

Frontend, backend, payments, playback, subscription access, progress persistence, debugging and privacy review

17 files in the primary completed change set	354 lines added	86 lines removed	207 automated tests passing
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Reporting date 18 September 2026

Repository Edu-nex_Final / Skillomate

Primary delivered commit f82289f - Fix subscription access and persistent course progress

Outcome. The web and backend suites are green, the production frontend build succeeds, local client/admin/API health checks return HTTP 200, and course progress now remains stable when lesson metadata becomes available.

Scope of this report

- Documents the work completed during the current Skillomate development and debugging cycle.
- Separates delivered changes from the latest uncommitted debugging refinements.
- Records verification evidence, known limitations and recommended next actions.

1. Executive summary

The work focused on stabilizing the paid learning experience across the browser, backend and mobile-compatible API. The largest completed change set corrected subscription entitlement behavior, made progress survive refreshes, improved payment simulation, hardened account deletion, and aligned the course-progress component with dark and light themes.

The follow-up debugging pass investigated the reported percentage decrease. The root cause was a denominator that changed as previously unknown lesson durations were discovered. Progress is now calculated as the average per-lesson coverage across the fixed lesson count, which prevents the displayed course percentage from dropping solely because metadata was filled later.

2. Delivered changes by area

Area	Completed work
Frontend / learner UI	Course progress now consumes the backend percentage; dark/light styling improved; learner subscription state remains accessible after refresh; admin controls were aligned with updated APIs.
Backend / access	Central subscription-access resolver preserves already-paid access until the actual period end, including cancelled subscriptions that should not renew.
Backend / progress	Progress writes, compatibility projections and certification synchronization were updated so the server is the durable source of truth.
Payments	The simulator supports the 24-hour trial workflow and idempotent behavior, reducing duplicate or inconsistent local subscription outcomes.
Account lifecycle	Deletion cleanup was expanded across dependent data while keeping the user deletion as the final step.
Playback	Private playback authorization and session checks were exercised; CloudFront playback still depends on valid backend key configuration.

Primary completed change-set size

Commit f82289f changed 17 files with 354 additions and 86 deletions across backend services, controllers, middleware, frontend components, admin code and automated tests.

3. Defects addressed

- Progress reset after refresh: persisted server progress is loaded instead of relying only on transient browser state.
- Percentage decreasing: percentage no longer shrinks when additional lesson durations become known.
- Subscription requested again locally: access resolution now respects an active trial or remaining paid period even when the raw gateway status is cancelled or stale.
- Payment simulator inconsistency: simulator responses and trial duration were normalized and tested for repeat requests.
- Theme mismatch: the course-progress panel has explicit dark/light presentation rules.
- Test regressions: backend harnesses were updated to provide current helpers used by playback, AI history, mobile player and session-refresh code.

4. Verification evidence

Check	Evidence	Result
Backend automated suite	145 / 145 passed	PASS
Frontend player suite	62 / 62 passed	PASS
Frontend production build	116 modules transformed	PASS
Mobile Expo configuration	Public configuration resolved successfully	PASS
Backend health endpoint	HTTP 200 on localhost:3000	PASS
Client application	HTTP 200 on localhost:5173	PASS
Admin route	HTTP 200 on localhost:5173/admin	PASS
Git whitespace validation	No diff-check errors	PASS

Build note: Vite reports one performance warning because the HLS chunk is larger than 500 kB after minification. This does not fail the build, but code splitting is a future optimization.

5. Current working-tree changes

The latest debugging refinements remain uncommitted and have not been pushed. They affect five files: one production backend rule and four automated-test files, totaling 18 additions and 3 deletions.

File	Purpose
edunex-b/services/completionRules.js	Stable average per-lesson percentage calculation.
edunex-b/tests/certification.test.cjs	Regression test for newly discovered lesson durations.
edunex-b/tests/ai-chat-history.test.cjs	Updated isolated test helper context.
edunex-b/tests/mobile-player.test.cjs	Updated course-loader test dependency.
edunex-b/tests/web-session-refresh.test.cjs	Updated revoked-session handler dependency.

6. Privacy and activity-tracking review

The application processes personally identifiable and behavioral information required for account, learning and support features. This includes names, mobile numbers, optional email, age/gender selections, avatar data, session/device context, learning progress, subscription information and support messages.

Current activity records cover login sessions, IP/user-agent context, video starts, playback progress, video and course completion, downloads, wishlists, payments, subscriptions, certificates and selected administrative actions. The system does not currently maintain a comprehensive clickstream for every button, link, page view, search, scroll or form abandonment.

Recommended boundary

Track meaningful product events, not raw keystrokes or indiscriminate clicks. Never place passwords, OTPs, payment-card data or sensitive message content in analytics payloads. Update consent, privacy and retention controls before expanding behavioral tracking.

7. Recommended next actions

- Review and commit the five current debugging changes after product-owner approval.
- Retest with a real subscribed local account to confirm entitlement, playback and progress behavior end to end.
- Configure private CloudFront playback with the correct backend private key and public key ID in the deployment environment.
- Remove tracked iOS build logs and archives from future source-control commits because they expose developer/build metadata and unnecessarily enlarge the repository.
- Define an analytics event catalog before adding click tracking, including purpose, fields, retention period and deletion behavior.
- Optimize the HLS bundle through lazy loading or manual chunking if initial-load performance becomes a concern.

Status: ready for review. Automated checks are green. The outstanding decision is whether to commit and push the latest five-file debugging refinement.

BUILD UPDATE - COURSE OWNERSHIP

Comparison of the previous subscription-only learner controls with the completed individual-course purchase and learner-dashboard workflow.

Previous behavior	New behavior
Admin-created learners received no access or broad subscription access.	Admins can assign a specific published course when creating a learner.
Learner controls offered Grant subscription and Remove access.	Learner controls offer Add purchased course and Remove course.
Protected course and playback routes checked subscription entitlement only.	Protected routes recognize ownership of the requested course; legacy subscriptions remain a compatibility fallback.

Notable code changes

- Added purchasedCourses references to learner records.
- Added an audited admin API for granting and revoking individual courses.
- Updated playlist, private playback and certification authorization to accept course ownership.
- Preserved subscription entitlement as a transition fallback for existing paying customers.
- Updated learner creation and management UI to use course selection instead of subscription duration.
- Replaced manual course-number prompts with an accessible in-page dropdown dialog and audit-reason field.
- Added per-course Trial (custom days), Yearly (365 days), and Permanent access with backend expiry enforcement.
- Added immediate purchased-course detail cards in learner management and synchronized the open learner drawer after ownership changes.
- Extended subscription-status responses with active course IDs and duration-aware entitlement details.
- Fixed the learner dashboard gate so an assigned course unlocks the panel without requiring a global subscription.
- Limited course-owner learning history to courses assigned by an administrator; subscribers retain catalog-wide access.

Verification

Backend: 8 targeted entitlement and playback tests passed, including legacy permanent grants and expired trial exclusion.

Frontend: 74 player tests passed and the production Vite build completed successfully.

Code quality: JavaScript syntax checks and git whitespace validation passed.

Deployment note

This build is implemented and verified locally. Production behavior will change only after the backend and frontend are deployed together.

BUILD UPDATE - LIVE USER PRESENCE

Admin learner management can now distinguish currently online learners from offline learners using authenticated session heartbeats.

Previous behavior	New behavior
Admin displayed historical last-active dates only.	Each learner displays Online or Offline with a colored presence dot.
Activity timestamps changed only through selected learning actions.	The learner app sends an authenticated heartbeat every 30 seconds while visible.
Admin data refreshed only after manual interaction.	Learner presence refreshes automatically every 30 seconds without replacing the page with a loading state.

Implementation

- Reused the authenticated session ping endpoint and persisted its latest timestamp.
- Defined Online as a heartbeat received within the previous three minutes, allowing for browser timer throttling.
- Continued browser heartbeats while the learner tab is in the background and sent an immediate ping on focus.
- Added server-calculated presence and last-seen data to admin learner-management responses.
- Added an Online Now summary card, row-level presence badges, and presence details in the learner drawer.
- Added a User presence filter with All users, Online now, and Offline options.
- Kept presence filtering separate from account access and subscription-status filters to prevent ambiguous Active labels.
- Renamed ambiguous UI values: Active subscription, Enabled Accounts, and User online status with Online users / Offline users.
- Fixed the learner Course progress record background and text colors so records remain readable in dark and light admin themes.

Verification

Backend: Presence boundary tests passed, including explicit background-tab throttling tolerance.

Frontend: 74 player tests passed and the production Vite build completed successfully.

Runtime: Updated backend restarted locally with MongoDB connected.

Operational behavior

A learner becomes Online after opening an authenticated user-panel page. If heartbeats stop, the learner changes to Offline after approximately three minutes. The admin screen updates within its next 30-second refresh.

BUILD UPDATE - LEARNER PURCHASE HISTORY

A dedicated Purchase history action now gives administrators a unified learner commerce and course-access timeline.

Previous behavior	New behavior
Manage showed course ownership and a technical admin-action list.	Every learner row now includes a Purchase history button beside View and Manage.
Payment orders and manual course assignments were separate.	One chronological panel combines payment orders with course grant and revoke events.
Manual course assignments could not be understood as purchase/access history.	Admin assignments show course title, access duration, status, reason context, and date.

Implementation

- Added a protected per-user purchase-history endpoint.
- Returns up to 100 recent payment orders and 100 recent course ownership changes.
- Added a responsive, theme-aware Purchase history panel that expands beneath the learner row like Manage.
- The button changes to Close history while its inline panel is open.
- Reorganized row actions into a two-column grid: View/Manage, full-width Purchase history, then destructive actions.
- Formatted Indian learner mobile numbers for display as +91,99999-99992 without changing stored values or search behavior.
- Reduced the sidebar-only wordmark size and reserved fixed space for the collapse control so the full Skillomate brand is not clipped.
- Shows payment type, gateway, payment instrument, amount, status, and timestamp.
- Shows admin course grants and revocations with course title and Trial, Yearly, or Permanent access type.
- Added empty, loading, and error states.

Verification

Backend: 17 admin, entitlement, playback, and presence tests passed.

Frontend: 74 player tests passed and the production Vite build completed successfully.

Code quality: Backend syntax and git whitespace validation passed.

Data note

A manually assigned course is labeled Admin assigned rather than displaying a false payment amount. Real payment orders retain their stored INR amount and gateway status.